Questions and MCQ

Step 4A: Understand role of capital structure on capital budgeting

"Capital Structure in a Perfect Market"

1) Which of the following statements is FALSE?
A) The relative proportions of debt, equity, and other securities that a firm has outstanding constitute its capital structure.
B) The most common choices for financing a new project are financing through equity alone and financing through a combination of debt and equity.
C) The NPV created by a project represents the value to the new investors of the firm.
D) When corporations raise funds from outside investors, they must choose which type of security to issue.
Answer: C
Explanation: The NPV created by the project represents the value to the existing shareholders of the firm.

Use the information for the question(s) below.

Consider a project with free cash flows in one year of $90,000 in a weak economy or $117,000 in a strong economy, with each outcome being equally likely. The initial investment required for the project is $80,000, and the project's cost of capital is 15%. The risk-free interest rate is 5%.

2) The NPV for this project is closest to:
A) $6250.
B) $14,100.
C) $10,000.
D) $18,600.
Answer: C
Explanation: $NPV = \frac{(.5)\$90{,}000 + (.5)\$117{,}000}{1.15} - \$80{,}000 = \$10{,}000$

3) Suppose that to raise the funds for the initial investment, the project is sold to investors as an all-equity firm. The equity holders will receive the cash flows of the project in one year. The market value of the unlevered equity for this project is closest to:
A) $94,100.
B) $90,000.
C) $86,250.
D) $98,600.
Answer: B
Explanation: $PV(\text{equity cash flows}) = \frac{(.5)\$90{,}000 + (.5)\$117{,}000}{1.15} = \$90{,}000$

4) Suppose that to raise the funds for the initial investment the firm borrows $80,000 at the risk-free rate, then the cash flow that equity holders will receive in one year in a weak economy is closest to:
A) $6000.
B) $10,000.
C) $0.
D) $33,000.
Answer: A
Explanation: $90,000 - $80,000(1.05) = $6000

Use the following information to answer the question(s) below.

Galt Industries has 50 million shares outstanding and a market capitalization of $1.25 billion. It also has $750 million in debt outstanding. Galt Industries has decided to delever the firm by issuing new equity and completely repaying all the outstanding debt. Assume perfect capital markets.

5) The number of shares that Galt must issue is closest to:
A) 15 million.
B) 25 million.
C) 30 million.
D) 40 million.
Answer: C
Explanation: Share price = $\frac{\$1250 \text{ million market cap}}{50 \text{ million shares}} = \$25$

Number of new shares = $\frac{\$750 \text{ million}}{\$25} = 30 \text{ million}$

6) Suppose you are a shareholder in Galt industries holding 600 shares, and you disagree with this decision to delever the firm. You can undo the effect of this decision by:
A) borrowing $6000 and buying 240 shares of stock.
B) selling 240 shares of stock and lending $6000.
C) borrowing $9000 and buying 360 shares of stock.
D) selling 360 shares of stock and lending $9000.
Answer: C
Explanation: Share price = $\frac{\$1250 \text{ million market cap}}{50 \text{ million shares}} = \$25 \rightarrow$ Value of equity = $25 × 600 = $15,000

Galt's pre-delevered Debt/Equity = $\frac{\$750}{\$1250} = .60 \rightarrow$ for every $1 equity need $0.60 debt, so you need to borrow 0.60 × $15,000 = $9000 and then buy $9000/$25 = 360 more shares of stock.

7) Which of the following statements is FALSE?
A) The Law of One Price implies that leverage will affect the total value of the firm under perfect capital market conditions.
B) In the absence of taxes or other transaction costs, the total cash flow paid out to all of a firm's security holders is equal to the total cash flow generated by the firm's assets.
C) With perfect capital markets, leverage merely changes the allocation of cash flows between debt and equity, without altering the total cash flows of the firm.
D) In a perfect capital market, the total value of a firm is equal to the market value of the total cash flows generated by its assets and is not affected by its choice of capital structure.
Answer: A
Explanation: The Law of One Price implies that leverage will not affect the total value of the firm under perfect capital market conditions.

8) Which of the following statements is FALSE?
A) Investors can alter the leverage choice of the firm to suit their personal tastes either by borrowing to reduce leverage or by holding bonds to add more leverage.
B) On the market value balance sheet the total value of all securities issued by the firm must equal the total value of the firm's assets.
C) The market value balance sheet captures the idea that value is created by a firm's choice of assets and investments.
D) One application of MM Proposition I is the useful device known as the market value balance sheet of the firm.
Answer: A
Explanation: Investors can alter the leverage choice of the firm to suit their personal tastes either by borrowing to increase leverage or by holding bonds to reduce leverage.

Use the information for the question(s) below.

Consider two firms: firm Without has no debt, and firm With has debt of $10,000 on which it pays interest of 5% per year. Both companies have identical projects that generate free cash flows of $1000 or $2000 each year. Suppose that there are no taxes, and after paying any interest on debt, both companies use all remaining free cash flows to pay dividends each year.

9) Fill in the table below showing the payments debt and equity holders of each firm will receive given each of the two possible levels of free cash flows:

	Without		**With**	
Free Cash Flow	Interest Payments	Equity Dividends	Interest Payments	Equity Dividends
1000				
2000				

Answer:

	Without		**With**	
Free Cash Flow	Interest Payments	Equity Dividends	Interest Payments	Equity Dividends
1000	0	1000	500	500

2000	0	2000	500	1500

10) Suppose that Rearden Metal currently has no debt and has an equity cost of capital of 12%. Rearden is considering borrowing funds at a cost of 6% and using these funds to repurchase existing shares of stock. Assume perfect capital markets. If Rearden borrows until they achieved a debt-to-equity ratio of 50%, then Rearden's levered cost of equity would be closest to:
A) 10.0%.
B) 12.0%.
C) 15.0%.
D) 16.0%.
Answer: C
Explanation: $re = ru + \frac{D}{E}(ru - rd) = 12\% + 50\%(12\% - 6\%) = 15\%$

11) Which of the following statements is FALSE?
A) If we can identify a comparison firm whose assets have the same risk as the project being evaluated, and if the comparison firm is levered, then we can use its equity cost of capital as the cost of capital for the project.
B) We can calculate the cost of capital of the firm's assets by computing the weighted average of the firm's equity and debt cost of capital, which we refer to as the firm's weighted average cost of capital (WACC).
C) The portfolio of a firm's equity and debt replicates the returns we would earn if the firm were unlevered.
D) When evaluating any potential investment project, we must use a discount rate that is appropriate given the risk of the project's free cash flow.
Answer: A
Explanation: If we can identify a comparison firm whose assets have the same risk as the project being evaluated, and if the comparison firm is levered, then we can use its unlevered equity cost of capital as the cost of capital for the project.

12) Which of the following equations would NOT be appropriate to use in a firm with risky debt?

A) $\beta E = \beta U + \frac{D}{E}(\beta U - \beta D)$

B) $\beta U = \beta E + \frac{D}{E}(\beta U - \beta D)$

C) $\beta E = \beta U + \frac{D}{E}\beta U$

D) $\beta U = \frac{E}{E + D}\beta E + \frac{D}{E + D}\beta D$

Answer: C

Use the information for the question(s) below.

Luther Industries has no debt, a total equity capitalization of $20 billion, and a beta of 1.8. Included in Luther's assets are $4 billion in cash and risk-free securities.

13) What is Luther's enterprise value?
A) $16 billion
B) $10.5 billion
C) $24 billion
D) $20 billion
Answer: A
Explanation: Enterprise value = market value - cash = $20 billion - $4 billion = $16 billion

Use the following information to answer the question(s) below.

Nielson Motors is currently an all-equity financed firm. It expects to generate EBIT of $20 million over the next year. Currently Nielson has 8 million shares outstanding and its stock is trading at $20.00 per share. Nielson is considering changing its capital structure by borrowing $50 million at an interest rate of 8% and using the proceeds to repurchase shares. Assume perfect capital markets.

14) Nielson's EPS if they choose not to change their capital structure is closest to:
A) $2.00.
B) $2.30.
C) $2.50.
D) $2.90.
Answer: C
Explanation: EPS = NI/shares outstanding = $20 million/8 million = $2.50 Note NI = EBIT in this case

15) Nielson's EPS if they change their capital structure is closest to:
A) $2.00.
B) $2.30.
C) $2.50.
D) $2.90.
Answer: D
Explanation: Nielson will repurchase $50 million/$20 per share = 2.5 million shares
This leaves 8 million - 2.5 million = 5.5 million shares outstanding
Net Income = EBIT - Interest expense (no taxes) = $20 million - $50 million × 8% = $16 million available to shareholders. EPS =NI/shares outstanding = $16 million/5.5 million = $2.91

Use the information for the question(s) below.

Assume that Rose Corporation's (RC) EBIT is not expected to grow in the future and that all earnings are paid out as dividends. RC is currently an all-equity firm. It expects to generate earnings before interest and taxes (EBIT) of $6 million over the next year. Currently RC has 5 million shares outstanding and its stock is trading for a price of $12.00 per share. RC is considering borrowing $12 million at a rate of 6% and using the proceeds to repurchase shares at the current price of $12.00.

16) Show mathematically that the stock price of RC won't change following the debt issuance and share repurchase.
Answer: *EPS* = (*EBIT*)/Shares outstanding = ($6M)/5M shares = $1.20 EPS (unlevered)

$$V = \frac{EPS}{rU}$$

$$\$12.00 = \frac{\$1.20}{rU} \text{ so } rU = .10$$

$$rE = rU + \frac{D}{E}(rU - rD)$$

$$rE = .10 + \frac{12}{60 - 12}(.10 - .06) = .11 \text{ or } 11\%$$

EPS = (*EBIT* - Interest)/Shares outstanding = ($6M - .06 × $12M)/4M shares = $1.32 *EPS*
$V = \frac{\$1.32}{.11} = \$12.00$ which equals the original stock price.